EE5178 : Assignment 2
Panorama Stitching & Single View Geometry

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1 Objective

The objective of this assignment is to implement a complete panorama stitching pipeline using feature matching, homography estimation and RANSAC. Additionally, we analyze the limitations of homography when applied to scenes with depth variation.

2 Part 1: Baseline Implementation

2.1 Q1: Feature Detection and Matching

We detect keypoints and descriptors using SIFT. Feature correspondences are obtained using a brute-force matcher along with Lowe's ratio test to remove ambiguous matches.

Output:

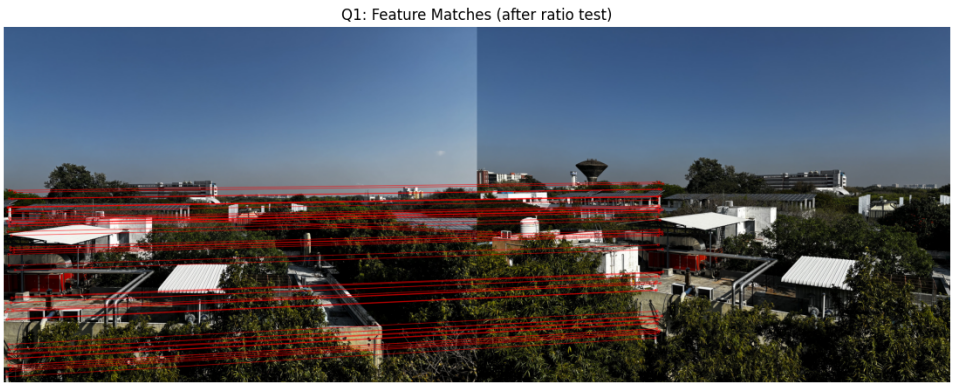

Figure 1: Feature matches between the two images

2.2 Q2: Homography using DLT

Given matched point correspondences, the homography matrix H is computed using the Direct Linear Transform (DLT). The system of equations $Ah = 0$ is solved using Singular Value Decomposition (SVD).

Homography Matrix:

```
[[-1.53319003e+00 -6.54098990e-01  1.41874114e+03]
 [-8.58909433e-01 -1.83352357e+00  1.42086243e+03]
 [-1.16000041e-03 -1.12702063e-03  1.00000000e+00]]
```

2.3 Q3: Robust Estimation using RANSAC

To handle outliers in feature matches, RANSAC is used. At each iteration, a random subset of correspondences is used to estimate H, and inliers are determined using reprojection error.

Output:

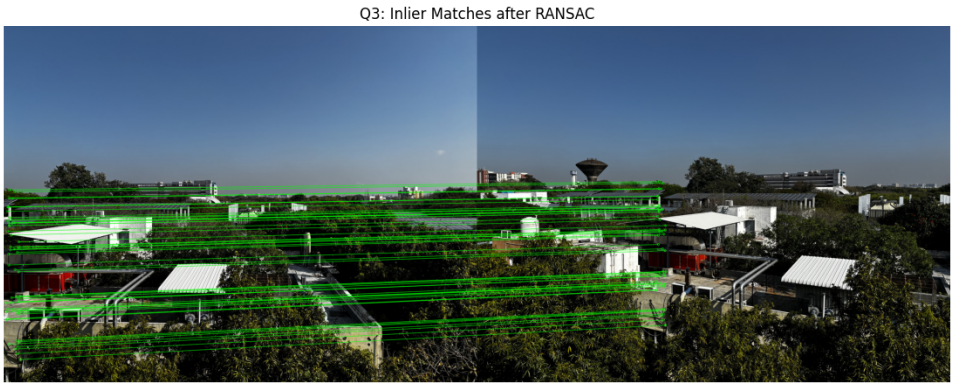

Figure 2: Inlier correspondences after RANSAC

2.4 Q4: Warping and Stitching

The second image is warped into the coordinate frame of the first image using the estimated homography. The final panorama is obtained by placing both images on a common canvas.

Output:

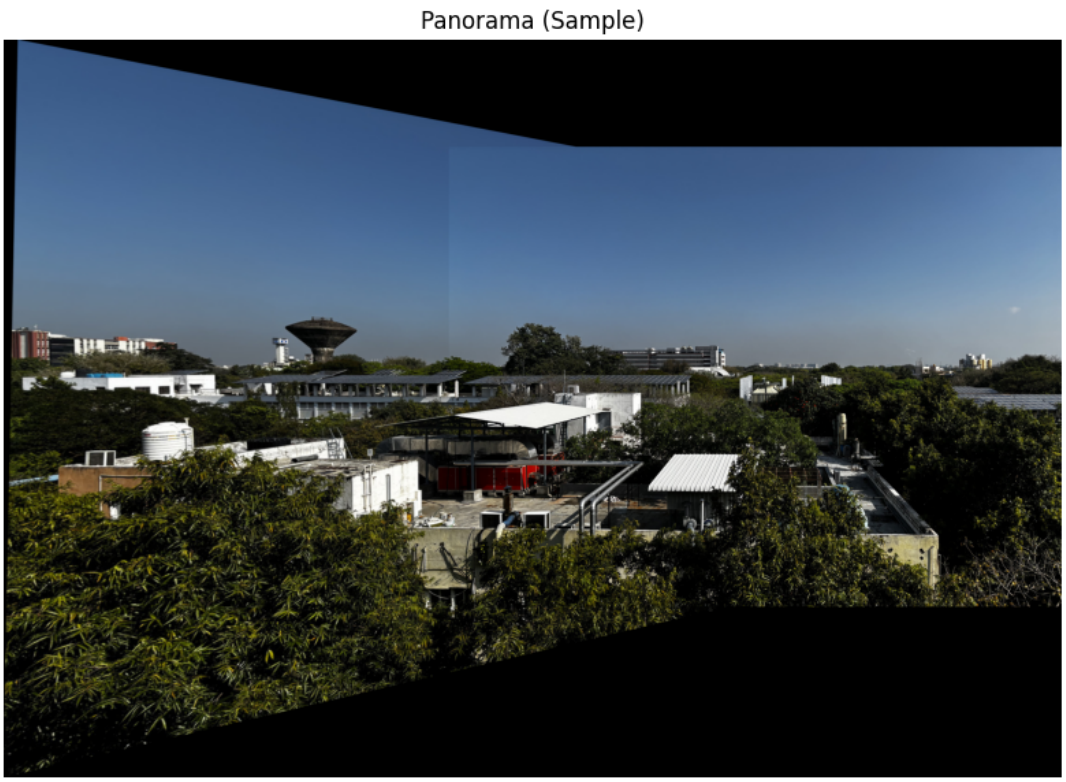

Figure 3: Panorama for sample image pair

3 Part 2: Data Collection in the Wild

3.1 Set A: Far Distance Scene

This dataset contains images of a distant scene where depth variation is minimal.

Two-image panorama:

Figure 4: Set A panorama using two images

Three-image panorama:

Figure 5: Set A panorama using three images

3.2 Set B: Near Distance Scene

This dataset contains images of a close scene with significant depth variation.

Two-image panorama:

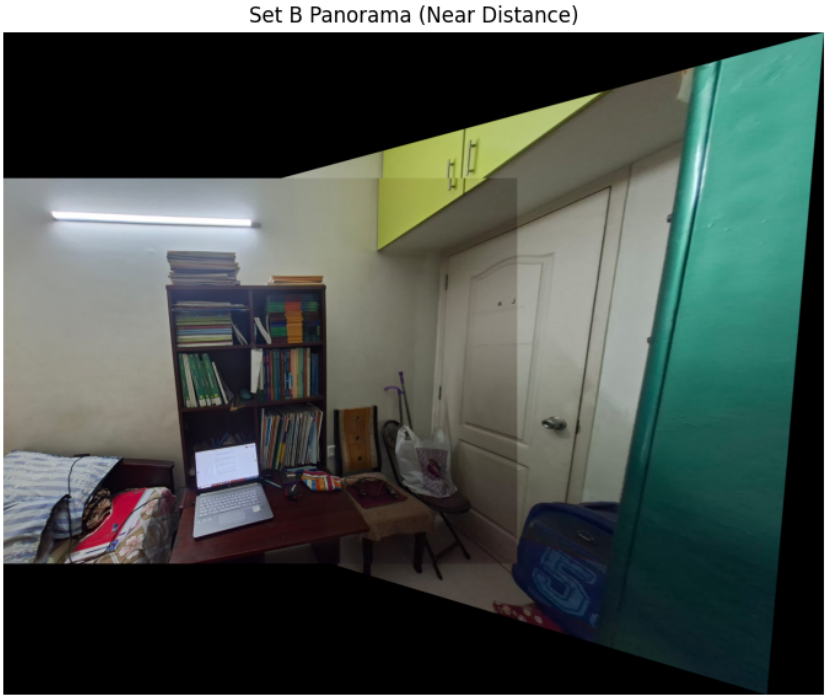

Figure 6: Set B panorama using two images

Three-image panorama:

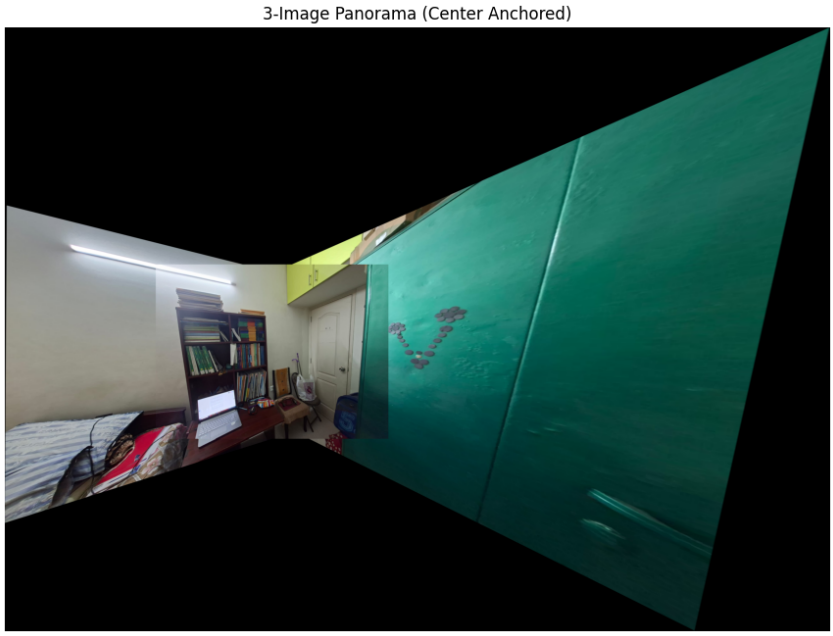

Figure 7: Set B panorama using three images

4 Part 3: Analysis

4.1 Observation

Set A produces a clean panorama with minimal visible seams. Set B shows ghosting and misalignment in overlapping regions. When extended to three images, these artifacts become more noticeable in Set B.

4.2 Geometric Justification

Homography assumes that the scene is planar or that the camera undergoes pure rotation.

The first scene consists of relatively distant houses, and hence the small amount of translation during capture does not significantly affect the alignment.

The second scene is an indoor setting, where objects are at varying depths. To capture a similar field of view, the camera undergoes both rotation and unintended translation.

In near-distance scenes, this translation leads to motion parallax, where points at different depths shift differently between images. This violates the homography assumption and results in visible misalignment.

4.3 Proposed Solution

To avoid these artifacts, the camera should be rotated about its optical center using a tripod. This ensures pure rotational motion, under which homography remains valid even for 3D scenes. In the current setup, handheld capture introduces unavoidable translation, which leads to the stitching errors observed above.

5 Conclusion

A complete panorama stitching pipeline was implemented successfully. The method works well for far-distance scenes but fails in near-distance scenarios due to violation of geometric assumptions.